



ARTIFICIAL INTELLIGENCE

AI is advancing at a great speed, and is making a great impact on automakers, vehicle owners and service providers in multiple ways. Many big players in the automotive industry are striving to automate their cars with AI-based technology. Many firms are also developing technology that can help with the installation of self-sufficient automobiles.

Self-driving cars

Japan's Tsukuba Mechanical Engineering Laboratory developed the first self-driving car in 1977. Since then, many companies have come up with their versions of self-driving cars. Use of public roads for testing self-driving cars has been legalised and permitted in many cities in the US, Europe and Asia.

In 2016, Uber began testing self-driving cars on the streets in San Francisco, USA. In March 2018, one of its self-driving cars—a modified Volvo XC90—that was speeding at about 64.4kmph (40mph) in autonomous mode hit a woman and killed her. This was the first major accident where a self-driving car killed a human being. Testing was temporarily put on hold for a few months, but it resumed again the same year.

In December 2018, Waymo became the first company to commercialise a fully-autonomous taxi service in the US.

A number of components contribute to the functioning of self-driving cars. These cars are built with complex systems such as braking, lane changing, collision prevention, navigation and mapping, which are controlled and monitored by high-performance computers or intelligent systems.

One such intelligent system is RT-RK computer system based on TI processors. RT-RK Alpha development board is used for advanced warning systems, active control systems and semi-autonomous vehicle applications.

Mapping is one of the most important functions in these vehicles. It handles pre-programmed data on approximations of streetlights, obstructions and surroundings of the areas being driven in and around. It also handles the safety of passengers during high-risk situations like head-on collisions with pedestrians. AI algorithms help perform various functions by processing and analysing input data, such as sounds and images from various sensors.

Some technologies used in AI are described below.

Cloud computing. This is an ideal platform for staging and deploying AI technology in automotive applications. It includes analytics, Big Data access and centralised connectivity. Cloud-based AI platforms help drivers pay for their fuel purchases from inside the vehicle, find nearby restaurants and buy household items from relevant stores. Applications of AI-cloud platforms ensure that data is available when needed.

The Internet of Things. There are many sensors connected to the Internet every day, from weather to healthcare to smartphones. The combination of AI and the Internet of Things (IoT) devices allow manufacturers to update vehicles with additional features such as predictive maintenance, repair scheduling, identification of performance issues, medical emergencies and automatic toll collection.

Machine learning. Machine learning, when applied to devices and machines, can think and act almost like human beings using past data. It can make informed deci-



Fig. 2: Uber's self-driving car, Volvo XC90 (Credit: <http://time.com>)

sions based on what it has learned. Big Data, machine learning and AI generate highly sensitive autonomous systems to support the movement of vehicles.

Deep learning. This is the process for achieving machine learning. It is based on an artificial neural network that can learn and make intelligent decisions on its own. Deep learning methods make a lot of sense with large data sets as compared to machine learning.

Deep learning helps in advanced driving assistance systems, and in enhancing the quality and characteristics of autonomous vehicles. Bosch is in the process of developing supercomputers for vehicles. Whereas, Nvidia is working to develop deep learning technology to power them.

Cognitive technology. Pattern identification, natural language interpretation, ability to build knowledge and cognitive analytics make sense of information. Cognitive computing can handle conceptual/symbolic data rather than just pure data. Cognitive systems work like a human and can interpret a real-life situation with unstructured data, making high-level decisions in complex situations. Car manufacturers have now started incorporating this technology into their vehicles. This is because self-driving involves more than following a set of rules or algorithms—it also involves learning.

AI does more than respond to what is happening in the vehicle's vicinity. Powerful deep learning algorithms can accurately predict what objects in the vehicle's path are likely to do. AI constantly learns and adjusts the rules to navigate the road. The result is a



Fig. 3: Waymo, the first commercial self-driving car (Credit: <https://en.wikipedia.org>)